Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech (CL) Examination

April 2018

CL207Concrete Technology

Date: 30.04.2018, Monday **Time: 10.00a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator and IS 10262 is allowed.

SECTION – I

Q.1 Answer The Following Questions

(i) What are the advantages of concrete over other materials? **[03]**

(ii) The aggregate in size smaller than 4.75mm is termed as **[01]**

(a) Coarse Aggregate **(b)**Fine Aggregate **(c)**All in one aggregate**(d)**none of the above

(iii) The lightweight aggregate can be prepared by **[01]**

(a)not using fine aggregate in concrete **(b)**using lightweight aggregate

(c)introducinga large quantity of air in concrete **(d)**all of the above methods.

(iv) The fiber reinforced concrete improves **[01]**

(a) Compressive strength **(b)**Tensile Strength **(c)**Durability **(d)**Workability

(v) The low heat cement is required when …………… is used. **[01]**

(a) Self compacting concrete **(b)**Mass concrete **(c)**Polymer concrete **(d)**High performance concrete

Q.2.a Attempt Any Three **[09]**

(i) Briefly summarize the cement manufacturing process by a wet method.

(ii) Define following

1) The standard consistency of cement 2) Initial setting time 3) Final setting time

(iii) Differentiate Rapid hardening cement and Quick setting cement.

(iv) What are the different types of mineral admixture? Explain any two in details.

Q.2.b Write aprocedure to find outthe compressive strength of cement inthelaboratory? **[05]**

OR

Q.2.b Discuss the field tests for the cement andprecautions required to store the cement at site. **[05]**

Q.3 Using IS 10262:2009 method of mix design; Calculatetheproportion of concrete for following data. **[14]**

1. A grade of concrete: M20
2. The degree of control: Good
3. The maximum size of aggregate: 20mm
4. Specific Gravity of Cement: 3.15
5. Specific Gravity of Fine Aggregate: 2.65
6. Specific Gravity of Coarse Aggregate: 2.7
7. The condition of Exposure: Mild
8. Workability:MediumSlump 25mm
9. Zone of Fine aggregate: Zone –II
10. Maximum free water-cement ratio is 0.55
11. Use Water Cement ratio: 0.47
12. Minimum Cement Content : 300kg/m^3

Assume if any additional data required.

SECTION – II

Q.4 **Answer The Following Questions** **[07]**

(i) Identify the instrument used for nondestructive test of concrete

(a) Compression testing machine **(b)** Universal testing machine

(b) Ultrasonic pulse velocity test **(d)** Both (b) and (c)

(ii) Use of Retarder in cement concrete

(a) Impart extreme workability **(b)** Reduces the amount of water to be added

(c) Improve bonding properties **(d)** Both (a) and (b)

(iii) pH value range of water for quality concrete shall be between

(a) 2-4**(b)**4-6**(c)**6-8**(d)**more than 8

(iv) The slump test is performed to identify the

(a) Strength **(b)**Workability**(c)**both a and b**(d)**None of the above

(v) Strength of concrete increase with increase in

(a) water/ cement ratio **(b)**size of aggregate **(c)**aggregate cement ratio

(d) cement/ water ratio

(vi) Workability of concrete can be found by

(a) Slump test **(b)** Compaction factor test**(c)**Vee Bee test **(d)** By all (a),(b) and (c)

(vii) Write different curing methods.

Q.5 Attempt Any Two **[14]**

a. Briefly, explain water-cement ratio and Gel space ratio and how itaffects the strength of concrete.

b. What are the tests of harden concrete? Explain any one in detail.

c. Differentiate between segregation and bleeding. What are the preventive measures for segregation and bleeding?

Q.6 Attempt Any Two **[14]**

a. What are the factors affecting workability of concrete? Discuss each in detail.

b. What do you understand by grading of aggregate? Write a stepwise procedure for sieve analysis.

c. State objectives of using air-entraining admixtures. What is a mechanism of air entraining agent?

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